

CHARAN SAI MACHA

Phone: +91 6301239837 Email: charansaimachal23@gmail.com LinkedIn GitHub

Summary

Motivated Computer Science undergraduate with hands-on experience building full-stack web applications using the MERN stack, RESTful APIs, and AI-integrated tools. Proficient in Java and Python with a strong foundation in Data Structures, Algorithms, and Object-Oriented Programming.

Education

Geethanjali College of Engineering and Technology, Hyderabad	<i>Oct 2022 – May 2026</i>
Bachelor of Technology in Computer Science and Engineering	GPA: 7.6
MJPTBC Welfare Junior College, Nagarjuna Sagar	<i>Jun 2020 – Mar 2022</i>
Intermediate (MPC)	Percentage: 95%

Technical Skills

- **Programming Languages:** Java (11+), Python, C, JavaScript, TypeScript
- **Frontend:** HTML, CSS/SCSS, React.js, Component Architecture, State Management
- **Backend:** Node.js, Express.js, Spring Boot, RESTful API Development, Microservices
- **Authentication & Security:** JWT, OAuth2, SSO, Role-Based Access Control
- **Databases:** MongoDB, MySQL, Oracle SQL, DBMS
- **Testing & QA:** Manual Testing, STLC, REST API Testing, Postman, HTTP Methods
- **Tools & Platforms:** Git, GitHub, VS Code, Vercel, Agile (Scrum/Kanban)
- **Concepts:** Data Structures and Algorithms, OOP, Concurrency, Multi-threading, Event-Driven Architecture

Certifications

- Deloitte Data Analytics Virtual Experience Program (Forage) *Jul 2025*
- IBM Python for Data Science Certification *Jun 2025*
- Oracle Certified – Java *Sep 2024*

Projects

Tasks Generator – AI-Driven Engineering Planning Tool

- Built an AI-driven full-stack planning tool that converts feature ideas into structured user stories and engineering tasks, following microservices-inspired modular architecture.
- Designed and implemented server-side RESTful API routes with Express.js; integrated @google/genai SDK for AI-powered backend processing.
- Implemented dynamic task editing, categorization, and multi-format export (TXT/Markdown) with responsive React.js frontend components and state management.
- Deployed production-ready application on Vercel with secure environment configuration and performance-optimized build pipeline.

Restaurant Reservation System (MERN Stack)

- Developed a scalable full-stack reservation system using MongoDB, Express.js, React.js, and Node.js following RESTful API design principles.
- Implemented JWT-based authentication and role-based access control (RBAC) for an Admin Dashboard, enforcing secure authorization workflows.
- Designed RESTful APIs with full CRUD operations, real-time reservation validation, and secure backend integration aligned with REST and HTTP standards.

Activities

- **Competitive Programming:** Solved 200+ DSA problems on LeetCode, HackerRank, and Codeforces, with focus on Java-based solutions covering concurrency, streams, and algorithmic optimization.